Almost Sharp Equivalence between Approximate Message Passing and Low-Degree Polynomials

Zhangsong Li

School of Mathemaitcal Sciences, Peking University

November 2026

High-dimensional statistical inference

- Find hidden structure in random graph:
  - planted clique;
  - stochastic block model.
- Find low-dimensional structure in random data:
  - spiked matrix model;
  - matrix/tensor decomposition.
- Regression/linear models:
  - compressed sensing;
  - phase retrieval.

Common features: large input, many unknowns, planted signal.

Spiked Wigner matrix model

$$\boldsymbol{Y} = \frac{1}{\sqrt{n}}\boldsymbol{\theta}\boldsymbol{\theta}^\top + \boldsymbol{W}$$

- Signal: $\theta_1, \ldots, \theta_n \overset{\text{iid}}{\sim} \pi_\Theta$, a known prior independent of n.
- Noise: W_{ij}, $1 \le i \le j \le n$, are independent $\mathcal{N}(0,1)$, and $\boldsymbol{W} = \boldsymbol{W}^\top$.
- Given $\boldsymbol{Y}$, estimate $\boldsymbol{\theta}$ under normalized mean-square error:

$$\frac{1}{n}\mathbb{E}\left\|\widehat{\boldsymbol{\theta}}(\boldsymbol{Y}) - \boldsymbol{\theta}\right\|_2^2.$$

- Simple "signal-plus-noise" model; natural testbed.

What are the best algorithms?

Statistical Physics

Approximate message passing

Belief Propagation

Free Energy Barrier

State Evolution

This work

Theoretical Computer Science

Low-degree polynomials

Spectral Methods

Sum-of-Squares

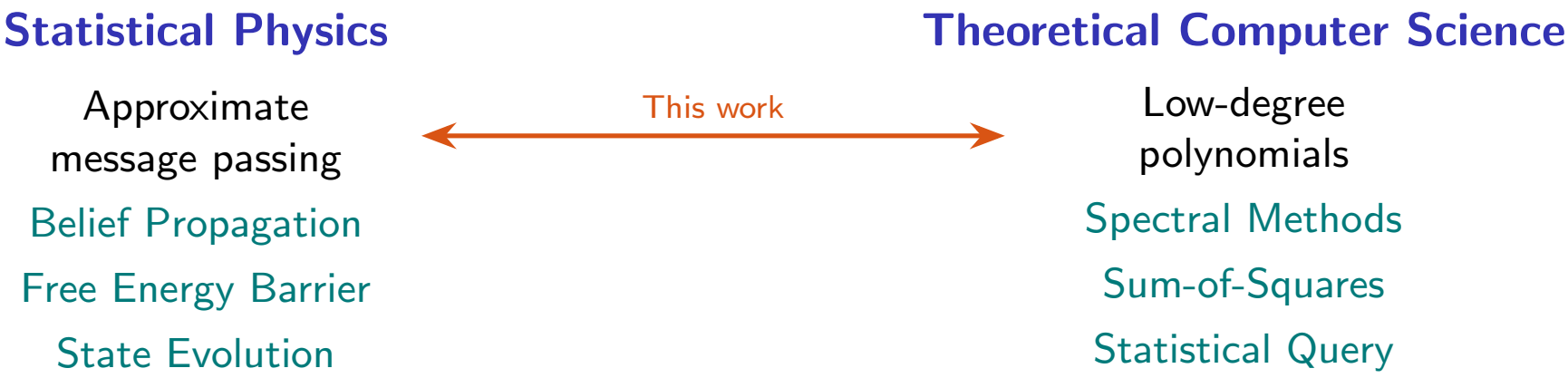
Statistical Query

What are the best algorithms?

Statistical Physics		**Theoretical Computer Science**
Approximate message passing	This work ⟷	Low-degree polynomials
Belief Propagation		Spectral Methods
Free Energy Barrier		Sum-of-Squares
State Evolution		Statistical Query

Unifying different perspectives?

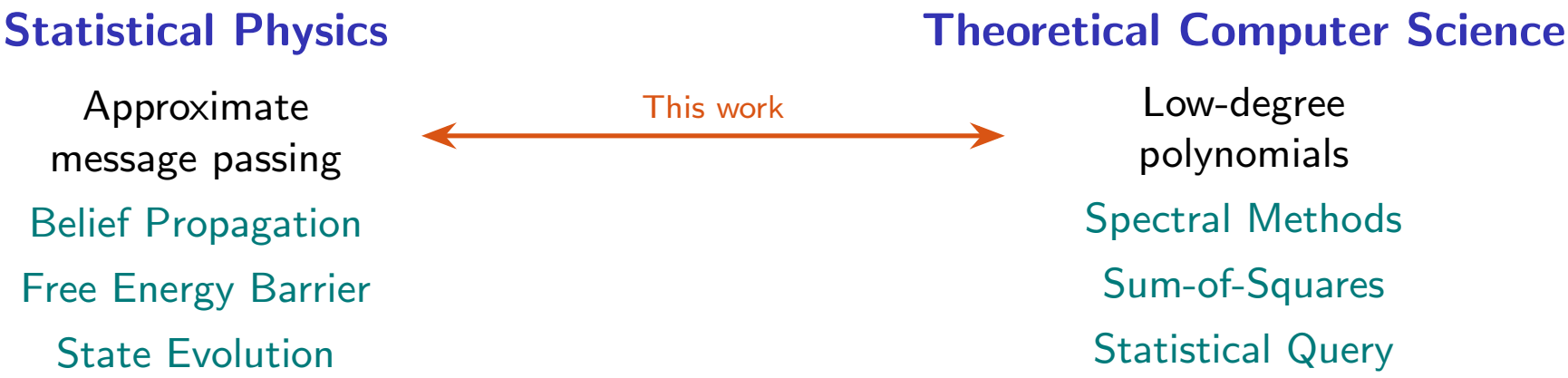
What are the best algorithms?



Unifying different perspectives?

- Not even answering the same question:
  - Statistical physics based prediction: optimization/sampling;
  - Algebraic based prediction: detection/refutation $\Longrightarrow$ recovery.

What are the best algorithms?



Unifying different perspectives?

- Not even answering the same question:
  - Statistical physics based prediction: optimization/sampling;
  - Algebraic based prediction: detection/refutation $\Longrightarrow$ recovery.
- Physics predictions are "wrong" for certain models (e.g., tensor PCA).

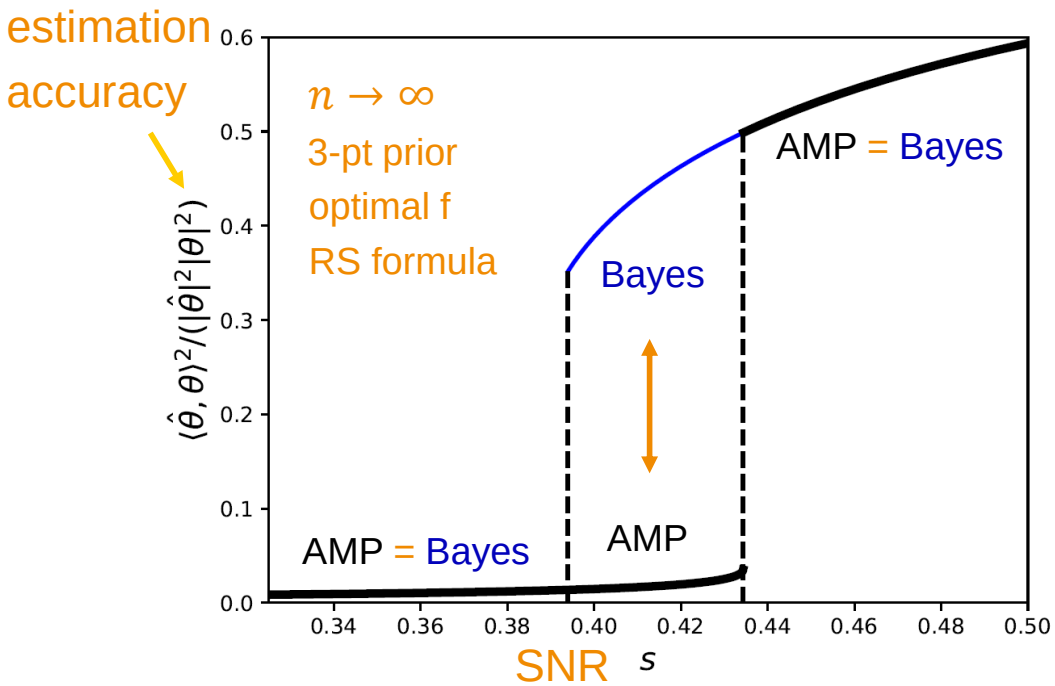
AMP benchmark

AMP alternates matrix multiplication with an entrywise denoiser:

$$\boldsymbol{x}^{t+1} = \frac{\boldsymbol{Y}}{\sqrt{n}} f_t(\boldsymbol{x}^t) - b_t f_{t-1}(\boldsymbol{x}^{t-1}).$$

b_t: Onsager correction

f_t: scalar denoiser



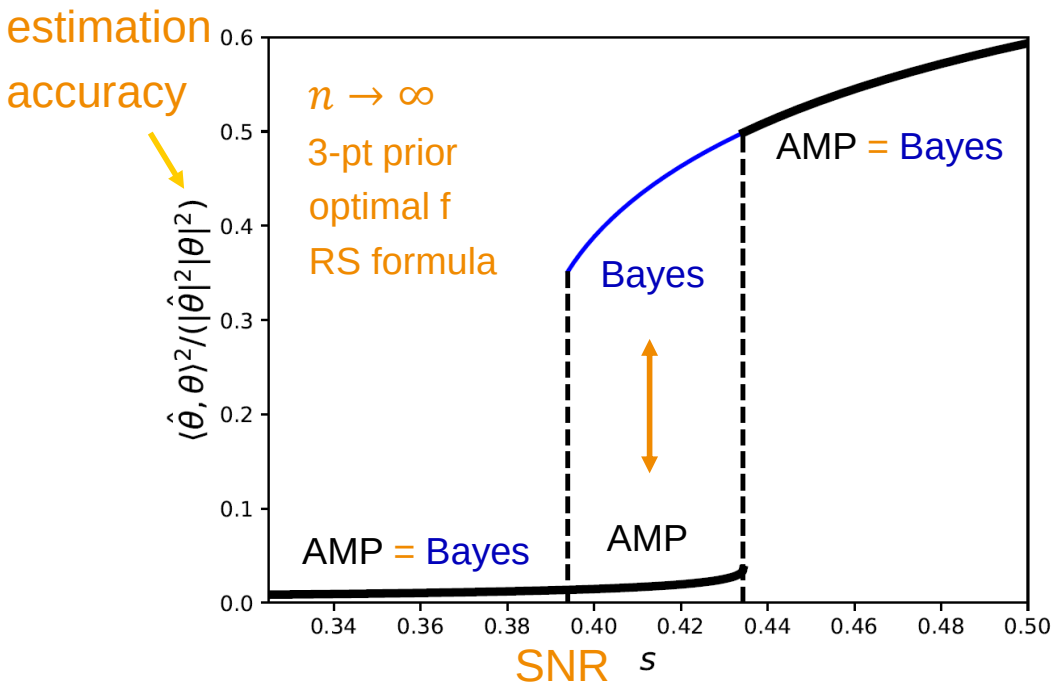
AMP benchmark

AMP alternates matrix multiplication with an entrywise denoiser:

$$\boldsymbol{x}^{t+1} = \frac{\boldsymbol{Y}}{\sqrt{n}} f_t(\boldsymbol{x}^t) - b_t f_{t-1}(\boldsymbol{x}^{t-1}).$$

b_t: Onsager correction

f_t: scalar denoiser



Bayes is not computational efficient $\Longrightarrow$ statistical-computational gap.

Main result

- Conjecture [Lesieur-Krzakala-Zdeborová 2015]: AMP has optimal MSE among all efficient algorithms.

Main result

- Conjecture [Lesieur-Krzakala-Zdeborová 2015]: AMP has optimal MSE among all efficient algorithms.
- AMP takes the form $(\widehat{\theta}_1, \ldots, \widehat{\theta}_n)$ is a const-deg multivariate polynomial in the entries of $\boldsymbol{Y}$.

Main result

- Conjecture [Lesieur-Krzakala-Zdeborová 2015]: AMP has optimal MSE among all efficient algorithms.
- AMP takes the form $(\widehat{\theta}_1, \ldots, \widehat{\theta}_n)$ is a const-deg multivariate polynomial in the entries of $\boldsymbol{Y}$.
- [Montanari-Wein 2025]: AMP has optimal MSE among all constant-degree polynomials.

Main result

- Conjecture [Lesieur-Krzakala-Zdeborová 2015]: AMP has optimal MSE among all efficient algorithms.
- AMP takes the form $(\widehat{\theta}_1, \ldots, \widehat{\theta}_n)$ is a const-deg multivariate polynomial in the entries of $\boldsymbol{Y}$.
- [Montanari-Wein 2025]: AMP has optimal MSE among all constant-degree polynomials.
- Our result: AMP has optimal MSE among all degree-n^{ϵ} polynomials.
  - [Hopkins 2018]: degree-D polynomials $\overset{\text{proxy}}{\Longleftrightarrow}$ algorithms with running time $n^{\widetilde{O}(D)}$.
  - Conjecture: the same holds all degree-$n^{1-\epsilon}$ polynomials.

Low-degree estimation lower bounds

- Given $\boldsymbol{Y}$, estimate $\boldsymbol{\theta}_1$.

Low-degree estimation lower bounds

- Given $\boldsymbol{Y}$, estimate $\boldsymbol{\theta}_1$.
- Reduce to understand $\mathrm{MMSE}_{\leq D} = \inf_{\deg(p) \leq D} \mathbb{E}[(p(\boldsymbol{Y}) - \theta_1)^2]$.

Low-degree estimation lower bounds

- Given $\boldsymbol{Y}$, estimate $\boldsymbol{\theta}_1$.
- Reduce to understand $\mathrm{MMSE}_{\leq D} = \inf_{\deg(p) \leq D} \mathbb{E}[(p(\boldsymbol{Y}) - \theta_1)^2]$.
- Various techniques have been developed to show $\mathrm{MMSE}_{\leq D} \to 1$. [Schramm-Wein 2022], [Luo-Zhang 2024], [Sohn-Wein 2025], [Even-Giraud-Verzelen 2026] ...

Low-degree estimation lower bounds

- Given $\boldsymbol{Y}$, estimate $\boldsymbol{\theta}_1$.
- Reduce to understand $\mathrm{MMSE}_{\leq D} = \inf_{\deg(p) \leq D} \mathbb{E}[(p(\boldsymbol{Y}) - \theta_1)^2]$.
- Various techniques have been developed to show $\mathrm{MMSE}_{\leq D} \to 1$. [Schramm-Wein 2022], [Luo-Zhang 2024], [Sohn-Wein 2025], [Even-Giraud-Verzelen 2026] ...
- Our work: exact value of $\lim_{n\to\infty} \mathrm{MMSE}_{\leq D}$ for $D = D(n)$.

Low-degree estimation lower bounds

- Given $\boldsymbol{Y}$, estimate $\boldsymbol{\theta}_1$.
- Reduce to understand $\mathrm{MMSE}_{\leq D} = \inf_{\deg(p) \leq D} \mathbb{E}[(p(\boldsymbol{Y}) - \theta_1)^2]$.
- Various techniques have been developed to show $\mathrm{MMSE}_{\leq D} \to 1$. [Schramm-Wein 2022], [Luo-Zhang 2024], [Sohn-Wein 2025], [Even-Giraud-Verzelen 2026] ...
- Our work: exact value of $\lim_{n\to\infty} \mathrm{MMSE}_{\leq D}$ for $D = D(n)$.
- Proof suggests how to obtain $\lim_{n\to\infty} \mathrm{MMSE}_{\leq D}$ for a problem where AMP is predicted to be optimal.

Proof sketch 1: modifying the Jensen approach

- Optimizing the scale of a scalar polynomial gives

$$\mathrm{MMSE}_{\leq D} = \mathbb{E}[\Theta^2] - \sup_{\deg(f)\leq D} \frac{\mathbb{E}[\theta_1 f(\boldsymbol{Y})]^2}{\mathbb{E}[f(\boldsymbol{Y})^2]}.$$

Proof sketch 1: modifying the Jensen approach

- Optimizing the scale of a scalar polynomial gives

$$\mathrm{MMSE}_{\le D} = \mathbb{E}[\Theta^2] - \sup_{\deg(f)\le D} \frac{\mathbb{E}[\theta_1 f(\boldsymbol{Y})]^2}{\mathbb{E}[f(\boldsymbol{Y})^2]}.$$

- Previous method [Schramm–Wein 2022], [Montanari-Wein 2025] begins with Jensen:

$$\mathbb{E}[f(\boldsymbol{Y})^2] \ge \mathbb{E}\big[\mathbb{E}[f(\boldsymbol{Y}) \mid \boldsymbol{W}]^2\big].$$

Proof sketch 1: modifying the Jensen approach

- Optimizing the scale of a scalar polynomial gives

$$\mathrm{MMSE}_{\le D} = \mathbb{E}[\Theta^2] - \sup_{\deg(f)\le D} \frac{\mathbb{E}[\theta_1 f(\boldsymbol{Y})]^2}{\mathbb{E}[f(\boldsymbol{Y})^2]}.$$

- Previous method [Schramm–Wein 2022], [Montanari-Wein 2025] begins with Jensen:

$$\mathbb{E}[f(\boldsymbol{Y})^2] \ge \mathbb{E}\big[\mathbb{E}[f(\boldsymbol{Y}) \mid \boldsymbol{W}]^2\big].$$

- Asymptotically tight only if revealing $\boldsymbol{\theta}$ does not affect much on f: only sharp when $\mathrm{MMSE}_{\le D} \to 1$.

Proof sketch 1: modifying the Jensen approach

- Optimizing the scale of a scalar polynomial gives

$$\mathrm{MMSE}_{\le D} = \mathbb{E}[\Theta^2] - \sup_{\deg(f)\le D} \frac{\mathbb{E}[\theta_1 f(\boldsymbol{Y})]^2}{\mathbb{E}[f(\boldsymbol{Y})^2]}.$$

- Previous method [Schramm–Wein 2022], [Montanari-Wein 2025] begins with Jensen:

$$\mathbb{E}[f(\boldsymbol{Y})^2] \ge \mathbb{E}\big[\mathbb{E}[f(\boldsymbol{Y}) \mid \boldsymbol{W}]^2\big].$$

- Asymptotically tight only if revealing $\boldsymbol{\theta}$ does not affect much on f: only sharp when $\mathrm{MMSE}_{\le D} \to 1$.

New idea

Retain the signal information needed at the AMP error level before applying Jensen.

Proof sketch 2: introduce auxiliary channel

- Use conditional Jensen:

$$\mathbb{E}[f(\boldsymbol{Y})^2] \geq \mathbb{E}\big[\mathbb{E}[f(\boldsymbol{Y}) \mid \mathcal{G}]^2\big].$$

Proof sketch 2: introduce auxiliary channel

- Use conditional Jensen:
$$\mathbb{E}[f(\boldsymbol{Y})^2] \geq \mathbb{E}\big[\mathbb{E}[f(\boldsymbol{Y}) \mid \mathcal{G}]^2\big].$$
- [Schramm-Wein 2022]: Direct Jensen bound $\mathrm{MMSE}_{\leq D}$ by cumulants $\mathrm{Cul}(\theta_1, (\theta_i\theta_j)^{[\alpha_{ij}]})$.
- [L. 2026]: Conditional Jensen bound $\mathrm{MMSE}_{\leq D}$ by conditional cumulants $\mathrm{Cul}(\theta_1, (\theta_i\theta_j)^{[\alpha_{ij}]} \mid \mathcal{G})$.

Proof sketch 2: introduce auxiliary channel

- Use conditional Jensen:
$$\mathbb{E}[f(\boldsymbol{Y})^2] \geq \mathbb{E}\big[\mathbb{E}[f(\boldsymbol{Y}) \mid \mathcal{G}]^2\big].$$
- [Schramm-Wein 2022]: Direct Jensen bound $\mathrm{MMSE}_{\leq D}$ by cumulants $\mathrm{Cul}(\theta_1, (\theta_i\theta_j)^{[\alpha_{ij}]})$.
- [L. 2026]: Conditional Jensen bound $\mathrm{MMSE}_{\leq D}$ by conditional cumulants $\mathrm{Cul}(\theta_1, (\theta_i\theta_j)^{[\alpha_{ij}]} \mid \mathcal{G})$.
- Introduce an auxiliary channel:
$$\boldsymbol{R} = \sqrt{s}\,\boldsymbol{\theta} + \boldsymbol{Z}, \quad s = \text{AMP fixed point}, \quad \boldsymbol{Z} \sim \mathcal{N}(0, I_n).$$

Proof sketch 2: introduce auxiliary channel

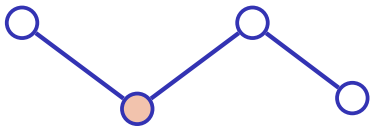
- Use conditional Jensen:
$$\mathbb{E}[f(\boldsymbol{Y})^2] \geq \mathbb{E}\big[\mathbb{E}[f(\boldsymbol{Y}) \mid \mathcal{G}]^2\big].$$
- [Schramm-Wein 2022]: Direct Jensen bound $\mathrm{MMSE}_{\leq D}$ by cumulants $\mathrm{Cul}(\theta_1, (\theta_i\theta_j)^{[\alpha_{ij}]})$.
- [L. 2026]: Conditional Jensen bound $\mathrm{MMSE}_{\leq D}$ by conditional cumulants $\mathrm{Cul}(\theta_1, (\theta_i\theta_j)^{[\alpha_{ij}]} \mid \mathcal{G})$.
- Introduce an auxiliary channel:
$$\boldsymbol{R} = \sqrt{s}\,\boldsymbol{\theta} + \boldsymbol{Z}, \quad s = \text{AMP fixed point}, \quad \boldsymbol{Z} \sim \mathcal{N}(0, I_n).$$
- Simple choice: $\mathcal{G} = \sigma(\boldsymbol{W}, \boldsymbol{R})$: bound $\mathrm{MMSE}_{\leq D}$ by $\boldsymbol{K}_\alpha(\boldsymbol{R}) = \mathrm{Cul}(\theta_1, (\theta_i\theta_j)^{[\alpha_{ij}]} \mid \boldsymbol{R})$.

Proof sketch 2: introduce auxiliary channel

- Use conditional Jensen:
$$\mathbb{E}[f(\boldsymbol{Y})^2] \geq \mathbb{E}\big[\mathbb{E}[f(\boldsymbol{Y}) \mid \mathcal{G}]^2\big].$$
- [Schramm-Wein 2022]: Direct Jensen bound $\mathrm{MMSE}_{\leq D}$ by cumulants $\mathrm{Cul}(\theta_1, (\theta_i\theta_j)^{[\alpha_{ij}]})$.
- [L. 2026]: Conditional Jensen bound $\mathrm{MMSE}_{\leq D}$ by conditional cumulants $\mathrm{Cul}(\theta_1, (\theta_i\theta_j)^{[\alpha_{ij}]} \mid \mathcal{G})$.
- Introduce an auxiliary channel:
$$\boldsymbol{R} = \sqrt{s}\,\boldsymbol{\theta} + \boldsymbol{Z}, \quad s = \text{AMP fixed point}, \quad \boldsymbol{Z} \sim \mathcal{N}(0, I_n).$$
- Simple choice: $\mathcal{G} = \sigma(\boldsymbol{W}, \boldsymbol{R})$: bound $\mathrm{MMSE}_{\leq D}$ by $\boldsymbol{K}_\alpha(\boldsymbol{R}) = \mathrm{Cul}(\theta_1, (\theta_i\theta_j)^{[\alpha_{ij}]} \mid \boldsymbol{R})$.
- Refine by averaging over $\boldsymbol{\theta}$: $\mathcal{G} = \sigma(\boldsymbol{W}, \{\boldsymbol{A}_\alpha(\boldsymbol{\theta})\})$, where $\boldsymbol{A}_\alpha(\boldsymbol{\theta}) = \mathbb{E}[\boldsymbol{K}_\alpha(\boldsymbol{R}) \mid \boldsymbol{\theta}]$.

Proof sketch 3: bounding the conditional cumulants

Rooted trees and $\alpha = 0$



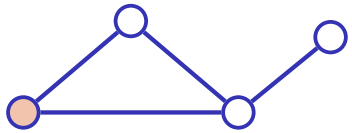
$\leq q_{\mathrm{AMP}}$

Disconnected or missing root



$= 0$

Rooted non-trees $g \geq 1$



$$\leq \sum_{g=1}^{D} \left(\frac{B_D}{n} \right)^g$$

Two crucial cases that the original cumulant bound loses control:

- Part 1: "tree part": use the fact that

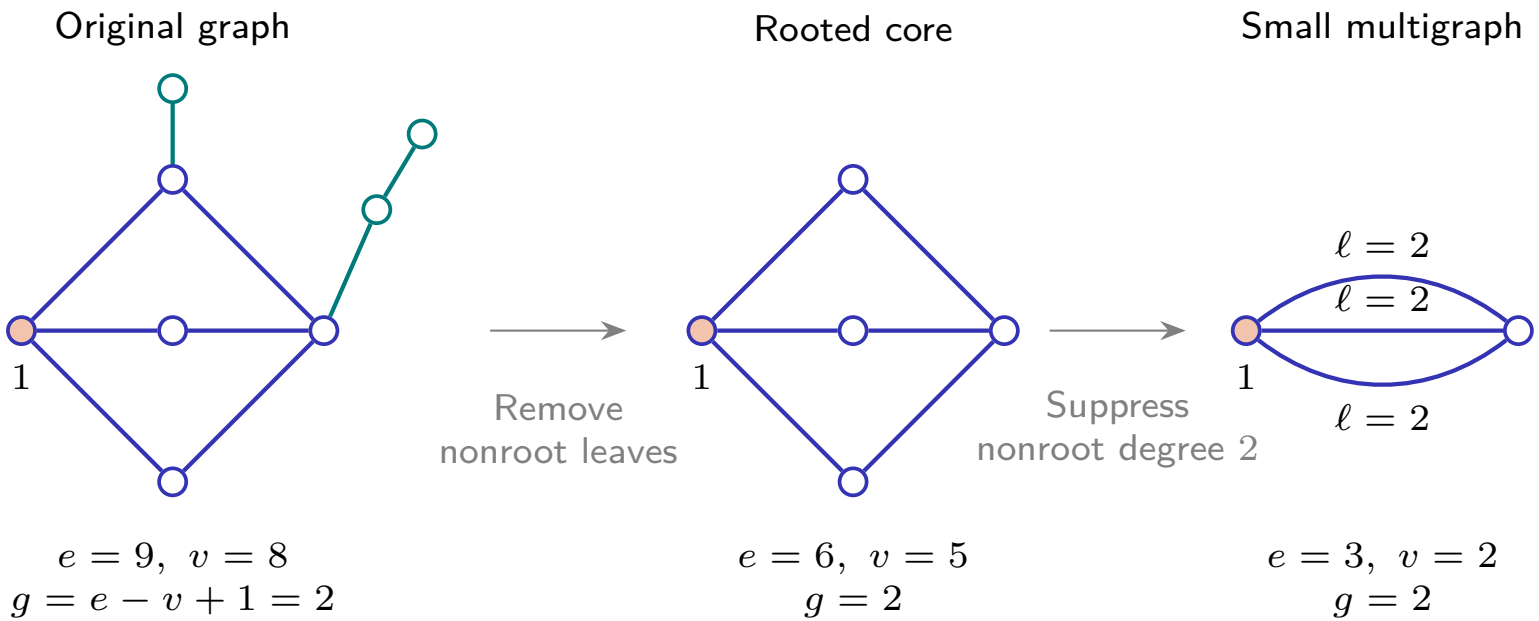
$$\mathbb{E}\big[\mathbb{E}[\theta_1 \mid \boldsymbol{R}]^2\big] = \text{AMP fixed point.}$$

  An iterated leaf-removing argument yields this part contributes roughly the same as AMP.

Proof sketch 3: bounding the conditional cumulants

Two crucial cases that the original cumulant bound loses control:

- Part 2: "non-tree part" with long chains:
  - Use the I-MMSE relation, the long chain yields to a exponentially decaying factor.
  - Reduce to the core graph contribution, which causes additional $n^{-\Omega(1)\cdot \text{graph excess}}$ loss.



Conclusion and future directions

Take-home message

For every fixed nonzero-mean sub-Gaussian prior, AMP achieves the asymptotically optimal MSE among all degree n^{ϵ} polynomials.

- Method: AMP-calibrated auxiliary channel, averaged conditional cumulants, and control by core excess.
- Open: larger degrees, mean-zero priors under sign-invariant loss, and other Gaussian additive models.

Thank you!